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Quantum Yield of the Ferrioxalate Actinometer

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The quantum yield of the potassium ferrioxalate actinometer at 365/6 mμ has been measured to be 1.26±0.03 relative to a calibrated thermopile. Comparisons of thermopiles were carried out at the Eppley Laboratory and at the National Bureau of Standards to ensure the flatness of thermopile response. Irradiations were made over the range 10^{12} to 3×10^{14} total absorbed photons.

INTRODUCTION

THE most thorough examination to date of the use of the photoreduction of potassium ferrioxalate as a chemical actinometer for the determination of fluxes of incident radiation has been made by Parker and coworkers,[1,2] revising the initial work of Allmand and Webb in 1929.[3] These former experimenters recommended a quantum yield of 1.21 at 365/6 mμ. In the course of our experiments on the measurement of the spectral sensitivity of electron multiplier phototubes,[4] we decided to use Parker's technique as an independent check of the photon fluxes previously measured using thermopiles calibrated, in our laboratory and elsewhere, by means of a National Bureau of Standards radiant energy standard lamp. The ferrioxalate actinometer, as reported by Parker,[2] is approximately one thousand times as sensitive as the older uranyl oxalate technique[5] and can be used easily in light fluxes of the same order of magnitude as are used in the calibration of our thermopiles. However, recently the uranyl oxalate system has been made even more sensitive than the ferrioxalate system by gas chromatographic determination of the photochemically produced carbon monoxide.[6]

In our preliminary experiments, we found that the photon fluxes determined with the ferrioxalate actinometer, using Hatchard and Parker's values of the quantum yield, differed significantly from those estimated with one of our calibrated thermopiles.[7] A more complete investigation has revealed two sources of error in actinometry measurements: First, metal-black coatings, such as gold black and platinum black, evaporated onto thermopile surfaces tend to deviate significantly from ideal blackness which we reported previously.[7] Second, we have found a reproducible, and as yet unexplained, enhancement of the quantum yield when the sides of the square fused silica cuvette are included in the irradiation beam.

EXPERIMENTAL

The preparations of materials and the procedures were followed according to the directions of Hatchard and Parker. Solutions were prepared in total darkness or in the presence of weak red illumination. The strong lines, 365/6 and 436 mμ from a Hg source, were used for excitations, although only the former was used extensively for the quantum yield determination. The yield at other wavelengths can be determined by relative comparison with the yield at 365/6 mμ.

The method is essentially a direct substitution technique, and therefore, should be independent of geometry or systematic error arising from nonhomogenous field of flux. Both a low-pressure "Osram" Hg discharge spectral lamp[8] and a collimated high-pressure "Osram" HBO 200 lamp were employed as sources with no difference in the results. The quantum yield reported here was determined with the low-pressure lamp.

After passing through an aperture, the beam traversed a filter consisting of two centimeters of a copper sulfate solution (10% weight/volume $CuSO_4\cdot5H_2O$ in 0.01N H_2SO_4). This solution transmits mainly in the region 300–650 mμ and, in combination with a glass filter (Corning C.S. 1–69), absorbs completely all wavelengths above 650 mμ. The photon fluxes were measured by means of a number of ac "Reeder" thermopiles[9] with the calibration light flux chopped at 10 cps, and also, with an "Eppley" dc thermopile, calibrated in our laboratory, with an NBS standard lamp, and also at the Eppley Laboratory.[10] The actinometer solution, at distances from 50 to 150 cm from the light sources, was always irradiated in a steady light flux. A glass color filter (Corning C.S. 3–73 for 365/6 mμ) was used to isolate the appropriate Hg line. Both the thermopile

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[7] Preliminary results were presented at the International Symposium on Photochemistry, University of Rochester, New York, March 1963 (unpublished).
[8] Ealing Corporation, Cambridge 40, Massachusetts.
[9] Charles M. Reeder and Company, 173 Victor Avenue, Detroit, Michigan.
[10] The Eppley Laboratory, Inc., Newport, Rhode Island.

detector and the actinometer solution were alternated in position immediately behind the filters, suitably shielded from light originating from outside the solid angle of the irradiation system. The stray radiation, as determined by the use of auxiliary, sharp cutoff, Corning filters, was much less than 1% of the incident intensity.

At first, the actinometer solution (3.0 ml) was contained in a 10-×10-mm fused silica fluorescence cuvette ("Pyrocell") with the entire front surface exposed to the radiation field. This technique gave rise to the enhancement of quantum yield. Later, this was modified by placing a small aperture of known area in front of the cuvette thus eliminating the sides from the direct light. Irradiations were made for periods ranging from 10–15 min in fluxes of $1\text{–}3\times10^{13}$ photons/sec-cm^2 (5–15 μW/cm^2 at 365/6 mμ). With these irradiations, sufficient ferrous ions were produced to give an optical density (510 mμ) of the ferrous–phenanthroline complex in the range 0.3–0.6. The optical density of the actinometer solution at 365 mμ was such that the incident radiation was completely absorbed. As the photochemically induced change in ferrioxalate concentration was only about 0.1%, nitrogen bubbling was found to be unnecessary. Unirradiated, but otherwise identical, blanks were run in matched cuvettes. The method of estimating ferrous ion produced was one of substitution. It was based on a titration with accurately standardized permanganate, and therefore, the requirement of the spectrophotometer operation was only that its response was linear in optical density. This eliminated systematic errors due to uncertainties in the absolute value of the optical density measurement. Nevertheless, our measured value of the extinction coefficient at 510 mμ of the ferrous–phenanthroline complex was within 0.5% of that previously reported.[2]

In checking experiments, it was found that the average beam intensity did not vary by more than $\pm1\%$ over periods of several hours. The areas of the thermopiles used (0.1 cm^2) were much smaller than the 1–10 cm^2 areas presented by the actinometer solution to the beam. However, the entire area was scanned by the thermopile, and it was established that the radiation field was uniform to within one percent. Corrections were made for light reflections at all surfaces. At the source-to-sample distances involved, refraction (lens) effects at the air-to-glass interface of the cuvette could be neglected.

RESULTS

The thermopiles used in the final measurements were known to be nonselective. The calibration was corrected for window absorption and, using a uniform, slightly diverging or parallel radiation flux of intensity $1\text{–}3\times10^{13}$ photons/sec·cm^2 incident on shielded areas of 0.1–10 cm^2, the following quantum yields of the potassium ferrioxalate system were measured for incident radiation of 365/6 mμ from a low-pressure Hg lamp:

0.006*M*

$K_3[Fe(C_2O_4)_3]$ 1.26±0.03 (17 determinations)

0.15*M*

$K_3[Fe(C_2O_4)_3]$ 1.20±0.03 (13 determinations).

These values are exactly the same as those obtained by Hatchard and Parker using a thermopile furnished by the National Physical Laboratory, although their recommended values, including lower results based on a comparison with the uranyl oxalate system, were 1.21 and 1.15, respectively.

DISCUSSION

A. Thermopile Blackness

The thermopile provides a method of measuring fluxes of radiation greater than about 0.1 μW/cm^2. Ideal operation of such a device would require that absorption be independent of the wavelength of the incident photons. Justification of this ideal "blackness," of course, will depend on the black coating applied to the thermopile surface, neglecting for the moment any effect of the thermopile window. These black coatings are usually thin deposits of lamp black or gold black. The optical properties of these types of coatings have been studied and show that the assumption of nonselectivity in the visible and near infrared regions should not lead to an error of more than a few tenths of one percent.[11] These results apply only to the bulk properties of the substances themselves, and not to an evaporated film on the thermopile junction surface. To make the thermopile response rapid the coating must be thin. If it is too thin, the backing material can be "seen" through it leading to spectral selectivity. It is not feasible, however, to determine the reflectivity of each thermopile, but, at least in the visible region, a good test is how black the coating looks to the eye; any visible coloration indicates selectivity.

Unexpectedly, the problem with the thin metal-black coating lies not in the visible, but in the infrared. Our thermopiles are calibrated by placing them in the field of an NBS radiant energy standard lamp.[12] As this is operated at a color temperature around 2300°K, most of the energy lies in the infrared between 1–3 μ. Therefore, if the calibrated thermopile is to be used to determine a flux of radiation in any other spectral region,

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there must be no difference in the reflectance of the black coating between this region and the 1–3 μ region. Our results show that some, but not all, gold-black coatings have a "hole" in the infrared, i.e., the reflectance is increased significantly, and when a thermopile having such a coating is calibrated, it yields light flux estimates for the visible region which are too high.

For a thermopile to have good responsivity, it must be of the vacuum type necessitating a window. The optical properties of this window must also be taken into account when the wavelength region of measurement is different from that used in the calibration. Quartz, for instance, is often used for thermopiles employed for ultraviolet measurement. It starts to absorb in the infrared at about 3 μ making a correction to the calibration necessary. This correction has been closely studied, and it was found, for a 1.5-mm-thick crystal quartz window, to amount to an absorption of about 8.5% for the light from an NBS radiant energy standard lamp operated at 0.35 or 0.40 A.[12] We calculated the absorption in other thicknesses of fused quartz by assuming that the extinction coefficient was the same as for crystal quartz and that Lambert's law was obeyed. The reflection from the two sides of the fused quartz window, amounting to a further 6%, must be added to this absorption to obtain the total radiant energy attenuation. The 6% reflectance correction for fused quartz in the infrared is essentially the same in the visible region.

Potassium bromide windows are also used on some of our thermopiles. There is no absorption correction to make to the calibration of a thermopile having one of these windows as the transmission is flat out to at least 15 μ. It should be observed, however, that the transmission starts to decrease below 400 mμ, and does so markedly by 300 mμ. The thermopile does not have a constant response across its sensitive area due to nonuniformity of coating, variable heat losses by conduction, etc. It is, therefore, essential to use and to calibrate the thermopile in identical radiation fields, a requirement which means that these should be uniform fields.

The divergence of our initial results from Hatchard and Parker's value, together with discussions with the Radiometry Section of NBS and the Eppley Laboratory, led us to suspect the ideality of the blackness of our gold-blacked thermopile surface. In independent measurements made at the National Physical Laboratory in England and the NBS and Eppley Laboratory in this country, both lamp black and a proprietary compound called "Parson's Black" have been found to be, essentially, ideally black from the ultraviolet through about 10 μ in the infrared. In reflectivity measurements, gold black has also been found to exhibit these flat characteristics. However, the difficulty with gold black apparently arises in the method of evaporation of a thin film of this substance that is required for high

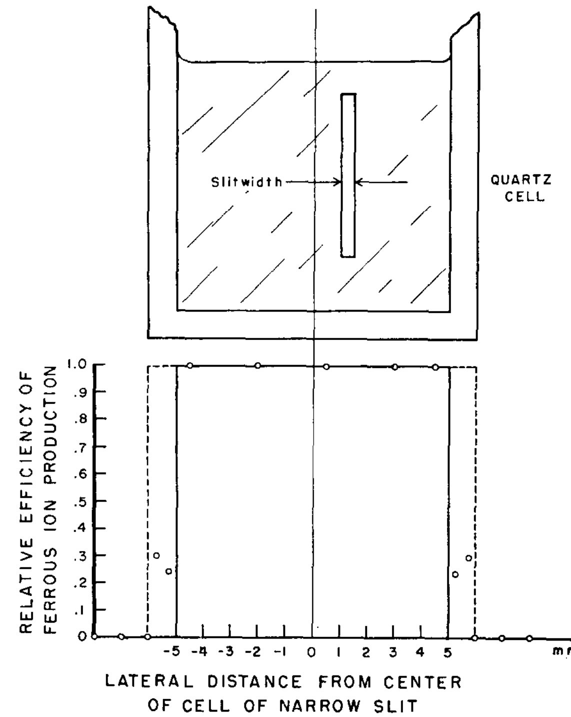

FIG. 1. Upper—scale drawing of the irradiation cell and slit dimensions. Lower—variation in photoreduction efficiency with slit position.

responsivity of the thermopile. Perhaps because lamp black and "Parson's black" surfaces are normally heavier than gold-black evaporations, they do not give rise to the same disability.

In a preliminary experiment, the facilities of the Eppley Laboratory were made available to us in order to compare our thermopiles with one of theirs which had been coated with "Parson's black." We used radiation greater than 700 mμ from an Hg source-filter combination, for normalization of the response of the thermopile in the infrared, for comparison of the "Parson's black" thermopile with our own at 365/6 mμ and 436 mμ. Since it had been shown that this "Parson's black" thermopile was equally black for all wavelengths, a relative comparison in this way should give an approximation to the selectivity. It was found that a 15% correction for lack of infrared blackness was required for one of our Reeder thermopiles and about 10% for our own Eppley thermopile. This latter thermopile was then recoated with lamp black to be used as a standard in our later blackness determinations. At this point, it should be emphasized that the reproducibility of thermopile calibrations between our laboratory and the Eppley Laboratory was ±0.5%.

We have found that the selectivity correction for a gold-blacked thermopile can vary from zero to 20%. Our precision was such that deviation from blackness greater than 1% could be measured. Only those thermo-

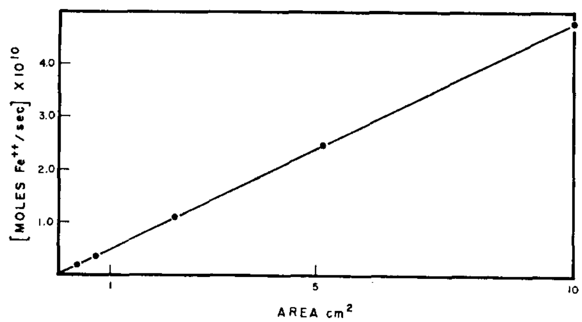

FIG. 2. Relative photoreduction as a function of area.

piles which were black to within $\pm 1\%$ were used for the quantum yield determinations reported in this paper.

B. Irradiation Area

In the preliminary measurements of the quantum yield the cuvette was used unmasked, although well shielded from stray light from any direction, and the irradiation area was assumed to be the area presented by the front surface of the cuvette containing the 3.0 cc of solution, i.e., 3.0 cm^2. High results, about 1.5, were obtained for the quantum yield. In the more detailed experiments, the cuvette was used in a holder with an aperture such that only a central portion, of total area about 1.0 cm^2, of the cuvette was irradiated. Experiments with a central portion of the cuvette exposed with varying width of mask, demonstrated a sudden increase in quantum yield as the edge of the cuvette was exposed to the flux. This effect is shown in Fig. 1 which is a plot of the derivative of results obtained as the aperture was narrowed from a total width larger than the cell to one including only a fraction of the solution portion of the cell. The ordinate is the relative efficiency of ferrous ion production. If there were no wall effect, the relative efficiency should fall to zero at the inner edge of the cuvette as shown by the solid dark line. If we assume that the area of the wall exposed to the beam is as effective as an equivalent area of solution in producing the photoreduction, the dashed curve of Fig. 1 would result. Our experimental points, as indicated by the black dots, are intermediate between these two extremes and indicate a surprising enhancement of photoreduction by light striking the edge of the cuvette. This edge effect cannot be explained by the possibility of fluorescence of the quartz sides of the cuvette as this cannot be expected to contribute more than a few percent. Diffraction effects and scattered light are also negligible.

The concentration of ferrioxalate ions in the bulk solution is 3.6×10^{18} molecules/cc. Even if we assume that a monomolecular layer of ions is adsorbed on the glass surface, and that, upon total reflection at the interface between the wall and the solution, the electric vector over one or two wavelengths of light is nonzero and can, therefore, interact with the adsorbed molecules,[13] the additional wall effect would amount to approximately a 1% increase in photoreduction. We can suggest no other mechanism to explain this effect.

Areas up to 10 cm^2 were irradiated in a Lucite cell. The path length was small in this case, 1.59 mm, and a more concentrated actinometer solution, 0.15M, was used in order to effect complete absorption at 365/6 mμ. Fig. 2 demonstrates a completely linear relationship between rate of Fe^{++} production and irradiation area for a constant intensity beam. In this case, a slightly lower quantum yield was obtained in agreement with Hatchard and Parker's data.

C. Sources of Error

The absolute accuracy of Hatchard and Parker's thermopile measurement is not known, but their precision appears to be of the order of 3%. No mention was made of the possibility of thermopile selectivity. However, their recommended quantum yield was 4% lower than their thermopile result, in order to take into account the results obtained by comparison with the uranyl oxalate actinometer. The recently improved sensitivity of the uranyl oxalate system should enable its quantum yield to be measured by the same method we have used for potassium ferrioxalate.

The present quantum yield measurement is subjected to the following sources of uncertainty:

(1) Uncertainty in calibration of the NBS radiant energy standard lamp including the uncertainty in the international radiation standard[12,14,15]	1%
(2) Thermopile calibration	1%
(3) Thermopile selectivity	0.5%
(4) Thermopile window absorption	0.5%
(5) Radiant energy estimation with thermopile including stray radiation effects	1%
(6) Estimation of total radiation incident on the solution considering cell wall reflection and absorption, and the area of irradiation	1%
(7) Spectrophotometric estimation of Fe^{++}	1%
(8) Standard deviation of quantum yield results	1%

rms uncertainty: $\pm 2.5\%$.

CONCLUSIONS

Our confirmation of Hatchard and Parker's result, using a calibrated thermopile for the quantum yield of

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the potassium ferrioxalate system under the conditions specified, leads us to recommend the value of 1.26±0.03 as a standard for actinometric work having an absolute accuracy indicated by the rms uncertainty. Owing to the uncertainties in the radiometry, it is not possible to reduce the error in quantum yield to below ±2% at this time.

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Effects of Anharmonicity on Vibrational Energy Transfer

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The diagonal matrix elements of the interaction potential which enter into the quantum theory of vibrational energy transfer generally have been *assumed* to be identical. Apparently the assumption is based on mathematic convenience rather than on any physical model. If Morse potentials are used to describe the intramolecular forces the elements are approximately but not identically equal. The transition probability is found to decrease markedly when the ratio of the diagonal elements of the initial and final oscillator states is allowed to deviate even slightly from one. Morse potentials were chosen to reproduce the observed anharmonicities of a variety of diatomic molecules and an anharmonic correction factor to the transition probability was calculated. It is found that the generally used calculated transition probabilities should be reduced by a factor of 10^{-1} to 10^{-2}.

I. INTRODUCTION

THE quantum theory of vibrational energy transfer in molecular collisions is founded on two major suppositions. The first is that the collision may be described by an adiabatic electronic interaction potential (and further it is assumed that the principal contributions to energy transfer come from the strongly repulsive portions of this potential where the largest changes in force during the course of a molecular vibration are expected). The second supposition is that since the observed efficiency of vibrational energy transfer is very small the dominant term in the wavefunction is the elastically scattered wave, and therefore only the coupling between the initial and final oscillator states need be considered. Together these assumptions dictate the use of the distorted wave approximation (DWA).

The DWA was first applied to the vibrational excitation of an oscillator in a head-on collision with a structureless incident particle by Jackson and Mott.[1] Many subsequent DWA calculations have been made, but primarily they are founded on this original work and represent further refinements of the physical model used to describe the collision.[2–8] Our concern in this paper is with a practice first introduced by Jackson and Mott, and used either explicitly or implicitly in all subsequent calculations, which we choose to call the "symmetric approximation." It consists of equating the diagonal matrix elements $W_{nn}=\langle n \mid W \mid n\rangle$ of the interaction potential W which enter into the DWA, i.e., $W_{jj}=W_{nn}$, where j and n refer to the initial and final oscillator states. Thus the "average" interaction is assumed to be independent of the quantum state of the oscillator, which implies that the potential is in a sense "vibrationally adiabatic." Principally this is introduced as a mathematical convenience and leads to a simple, closed-form expression for the transition

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